

Assignment No-1

04/04/26

Q.1. Define thresholding in image processing. Describe different types of thresholding techniques with examples.

Q.2. Differentiate between sampling and quantization. Explain with suitable diagrams or examples.

Q.3. A 3×3 grayscale image neighbourhood is given as:

$$\begin{bmatrix} 10 & 20 & 30 \\ 20 & 40 & 60 \\ 30 & 60 & 90 \end{bmatrix}$$

Filter mask is

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Find the smoothed output pixel value.

Q.4. A 3×3 grayscale image neighbourhood is given as:

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 50 & 50 \\ 10 & 50 & 50 \end{bmatrix}$$

Horizontal (Gx):

$$\begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

Vertical (Gy):

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix}$$

Find the Sobel edge magnitude at the centre pixel.

Q.5. Explain image intensity transformation techniques used in digital image processing. Describe the following methods with suitable mathematical expressions and examples:

1. Contrast Stretching
2. Log Transformation

3. Power-Law (Gamma) Transformation

- Q.6. Define image segmentation and explain its importance.
- Q.7. What is thresholding? Why is it called a similarity-based segmentation technique?
- Q.8. Differentiate between global thresholding and local thresholding.
- Q.9. What is region-based segmentation?
- Q.10. Define point detection in image processing.
- Q.11. What is an edge in a digital image?
- Q.12. Define line detection and mention its applications.
- Q.13. What is the hit-or-miss transform?
- Q.14. What is a structuring element in morphological operations?
- Q.15. List different types of discontinuities in images (point, line, edge).

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